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Ferroelectric tunnel junctions with conductive electrodes having asymmetric nitrogen or oxygen profiles

Abstract

A semiconductor device includes a ferroelectric tunnel junction (FTJ), wherein the ferroelectric tunnel junction includes a first electrode, a ferroelectric layer disposed over the first electrode, and a second electrode disposed over the ferroelectric layer. The first electrode contains nitrogen or oxygen and is characterized by a first percentage of nitrogen or oxygen. The second electrode contains nitrogen or oxygen and is characterized by a second percentage of nitrogen or oxygen. The first percentage is different from the second percentage.

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Background/Summary

BACKGROUND

(1) Volatile memory stores data while it is powered, while non-volatile memory is able to store data when power is removed. A ferroelectric random-access memory (FeRAM) device is an attractive non-volatile memory technology. This is because FeRAM devices provide many advantages, including a fast write time, high endurance, low power consumption, and low susceptibility to damage from radiation.

(2) FeRAM devices are based on ferroelectric material, which is characterized by a nonlinear relationship between the applied electric field and the stored charge. Specifically, the ferroelectric characteristic has the form of a hysteresis loop. Semi-permanent electric dipoles are formed in the crystal structure of the ferroelectric material. When an external electric field is applied across the ferroelectric material, the dipoles tend to align themselves with the field direction, produced by small shifts in the positions of atoms and shifts in the distributions of electronic charge in the crystal structure. When the external electric field is removed, the dipoles of the ferroelectric material retain their polarization state.

(3) Even though FeRAM devices are being commercialized, much improvement is desired.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1A is a cross-sectional view illustrating a ferroelectric tunnel junction that can be included in a memory cell of a ferroelectric memory device, in accordance with some embodiments.

- (3) FIG. 1B illustrates a ferroelectric hysteresis loop showing the relationship between the polarization (P) versus electric field (E) of a ferroelectric material.
- (4) FIG. 2 is a cross-sectional view illustrating a ferroelectric memory device which is a metal-ferroelectric-insulator-semiconductor (MFIS) ferroelectric memory device, in accordance with some embodiments.
- (5) FIG. 3 is a cross-sectional view illustrating a ferroelectric memory device **300** which is a 1 transistor-1 capacitor (1T1C) type ferroelectric memory device, in accordance with some embodiments.
- (6) FIG. 4 illustrates a ferroelectric tunnel junction and its band diagrams, in accordance with some embodiments.
- (7) FIG. 5 illustrates another ferroelectric tunnel junction and its band diagrams, in accordance with some embodiments.
- (8) FIG. 6 illustrates another ferroelectric tunnel junction and its band diagram, in accordance with some embodiments.
- (9) FIG. 7 illustrates another ferroelectric tunnel junction and its band diagram, in accordance with some embodiments.
- (10) FIG. 8 is a flowchart illustrating a method for forming a semiconductor device, in accordance with some embodiments.
- (11) FIG. 9 illustrates another ferroelectric tunnel junction and its band diagram, in accordance with some embodiments.
- (12) FIG. 10 is a table illustrating possible oxygen profiles in a ferroelectric tunnel junction, depending on the location of the interfacial layer, in accordance with some embodiments.
- (13) FIG. 11 illustrates another ferroelectric tunnel junction and its band diagram, in accordance with some embodiments.
- (14) FIG. 12 illustrates another ferroelectric tunnel junction and its band diagram, in accordance with some embodiments.
- (15) FIG. 13 illustrates another ferroelectric tunnel junction and its band diagram, in accordance with some embodiments.
- (16) FIG. 14 illustrates another ferroelectric tunnel junction and its band diagrams, in accordance with some embodiments.
- (17) FIG. 15 is a flowchart illustrating a method for forming a semiconductor device, in accordance with some embodiments.

DETAILED DESCRIPTION

(18) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(19) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted

accordingly.

(20) Typically, a ferroelectric memory device uses a ferroelectric layer, which is composed of a ferroelectric material. Various aspects of device performance (e.g., switching voltage, retention, endurance, etc.) of the ferroelectric memory device are highly dependent on the structure of the ferroelectric tunnel junction. Various embodiments of the disclosure are directed to provide ferroelectric memory devices with enhanced device performance.

(21) Embodiments of the disclosure provide a semiconductor device including a ferroelectric tunnel junction (FTJ), wherein the ferroelectric tunnel junction includes a first electrode, a ferroelectric layer disposed over the first electrode, and a second electrode disposed over the ferroelectric layer. The first electrode contains nitrogen or oxygen and is characterized by a first percentage of nitrogen or oxygen. The second electrode contains nitrogen or oxygen and is characterized by a second percentage of nitrogen or oxygen. The first percentage is different from the second percentage. The percentage of nitrogen or oxygen is adjusted to enhance the device performance of the ferroelectric tunnel junction. In some embodiments, varying the percentage of nitrogen in nitrogen-containing electrodes can lead to a different thickness of interfacial layers adjacent to each electrode, resulting in band bending that enhances tunneling current. In some other embodiments, varying the percentage of oxygen in oxygen-containing electrodes can lead to different work functions of each electrode, resulting in enhancement of tunneling current.

(22) FIG. 1A is a cross-sectional view illustrating a ferroelectric tunnel junction that can be included in a memory cell of a ferroelectric memory device, in accordance with some embodiments. Referring to FIG. 1A, in some embodiments, the ferroelectric tunnel junction **100** is formed on a substrate **101** and includes a ferroelectric layer **130** disposed between a first conductive region **110** and a second conductive region **120**. In some embodiments, the first conductive region **110** is a channel region of a transistor and the second conductive region **120** is a gate electrode of the transistor. In some other embodiments, the first conductive region **110** is a lower electrode connected to a transistor, and the second conductive region **120** is an upper electrode over the lower electrode. In some cases, the lower electrode is also referred to as the bottom electrode (BE), and the upper electrode is also referred to as the top electrode (TE). In some embodiments, other components (e.g., dielectric features) may be disposed between the ferroelectric layer **130** and the first conductive region **110**, and/or between the ferroelectric layer **130** and the second conductive region **120**. The ferroelectric layer **130** includes one or more of ferroelectric materials.

(23) FIG. 1B illustrates a ferroelectric hysteresis loop showing the relationship between the polarization (P) versus electric field (E) of a ferroelectric material. Referring to FIG. 1B, as the electric field increased from zero to a positive value, the overall polarization of the ferroelectric material increases as the polarization in different dipolar regions (domains) are aligned with the electric field. Eventually, the total polarization of the field reaches a saturation point where the polarization is not further increased as the electric field increases since all domains are aligned in the same direction. The saturation point is called the saturation polarization (P_s) of the ferroelectric material. When the electric field is reduced to zero (i.e., the electric field is removed), the ferroelectric material retains polarized, and the polarization value at this point is called the remnant polarization (P_r). A negative field will cause the polarization to reduce, until it reaches zero at the coercive field ($-E_c$). A further negative increase in the electric field will eventually cause a reverse saturation polarization ($-P_s$) to develop. When the electric field returns to zero, the ferroelectric material is left with a negative remanent polarization ($-P_r$). Increasing the field once more increases the polarization from $-P_r$ to zero at coercive field E_c , and then to P_s .

(24) Different ferroelectric materials have different properties. For example, the properties may include one or more of the remanent polarization, saturation polarization, coercive field, loop squareness, voltage-pulse time, grain size, the interface property and/or other electrical properties and film growth properties. Herein, the term “loop squareness” refers to the degree of squareness of

the hysteresis loop of the ferroelectric material. If the shape of the hysteresis loop is more like a square, the higher the loop squareness. "Voltage-pulse time" refers to the relationship between applied write pulse and required voltage to enable switching.

(25) In some embodiments, the ferroelectric layer **130** in FIG. **1** includes one or more ferroelectric materials selected from hafnium oxide (HfOx) doped with dopant(s) such as Zr, Si, La, AlScN, ZrOx, ZrOxPb3Ge5O11 (PGO), lead zirconatetitanate (PZT), SrBi2Ta2O9 (SBT or SBTO), SrB4O7 (SBO), SrBiBtacNbO4, (SBTN), SrTiO3 (STO), BaTiO3 (BTO), (BixLay)Ti3O12 (BLT), LaNiO3 (LNO), YMnO3, ZrO2, zirconium silicate, ZrAlSiO, HfO2, hafnium silicate, HfAlO, LaAlO, lanthanum oxide, Ta2O5, and/or other suitable ferroelectric material.

(26) In some embodiments, the ferroelectric layer **130** may include hafnium oxide (HfOx) doped with different dopants. The dopants may include various metallic dopants and/or semiconductor dopants. In some embodiments, the dopants may be selected from two or more of La, Zr, Si, Al, or the like and/or other suitable dopants.

(27) Interface property of a ferroelectric material refers to the property of the interface between the ferroelectric material and adjacent material. For example, when a ferroelectric material is disposed adjacent to a metal material, some ferroelectric materials may readily react with the metal material under thermal treatment or the metal is easy to diffuse into the ferroelectric material, which may negatively affect the electrical property of the ferroelectric material. In contrast, some ferroelectric materials do not readily react with metal material under thermal treatment and the metal material is not easy to diffuse into the ferroelectric material.

(28) On the other hand, when a ferroelectric material is disposed adjacent to an insulating material (such as native oxide), if the insulating material readily generates a depolarization field under the affection of the ferroelectric material, it represents the ferroelectric material has poor interface property. In contrast, if the insulating material does not readily generate a depolarization field when interfacing with or adjacent to the ferroelectric material, it represents the ferroelectric material has good interface property.

(29) The ferroelectric materials may be formed by suitable deposition processes, such as chemical vapor deposition (CVD), atomic layer deposition (ALD), physical vapor deposition (PVD), or the like or combinations thereof.

(30) The materials of the lower electrode **110** and upper electrode **120** may be respectively selected from a group consisting of aluminum (Al), titanium (Ti), copper (Cu), tungsten (W), platinum (Pt), palladium (Pd), osmium (Os), ruthenium (Ru), tantalum (Ta), or an alloy thereof, titanium nitride (TiN), tantalum nitride (TaN), tungsten nitride (WN), molybdenum nitride (MoN), TaSiN, TiSiN, WSiN, tungsten silicide, titanium silicide, cobalt silicide, zirconium silicide, platinum silicide, molybdenum silicide, copper silicide, nickel silicide, indium tin oxide (ITO), iridium oxide (IrO.sub.2), rhenium oxide (ReO.sub.2), rhenium trioxide (ReO.sub.3), or a combination thereof. The electrodes may be formed by suitable deposition processes, such as chemical vapor deposition (CVD), atomic layer deposition (ALD), physical vapor deposition (PVD), or the like or combinations thereof.

(31) The ferroelectric tunnel junction **100** may be applied in various types of ferroelectric memory devices. FIGS. **2** and **3** are schematic cross-sectional views illustrating memory devices including the ferroelectric tunnel junction **100** according to some embodiments of the disclosure.

(32) FIG. **2** is a cross-sectional view illustrating a ferroelectric memory device **200** which is a metal-ferroelectric-insulator-semiconductor (MFIS) ferroelectric memory device, in accordance with some embodiments. In some embodiments, the ferroelectric memory device **200** is a transistor, and a ferroelectric layer **230** is embedded in the transistor **200**. The transistor **200** may also be referred to as ferroelectric field effect transistor (FeFET).

(33) In some embodiments, the ferroelectric memory device **200** includes a substrate **201**. The transistor **200** includes an insulating layer **202**, a ferroelectric layer **230**, and a gate electrode **220** stacked on the substrate **201**, and source/drain (S/D) regions **205A** and **205B**. Source/drain (S/D)

regions **205A** and **205B** are disposed in the substrate **201** and adjacent to a channel region **210**. A channel region **210** is formed between the S/D regions **205A** and **205B** and underlying the gate electrode **220** and ferroelectric structure **230**.

(34) In some embodiments, the substrate **201** is made of silicon and/or other semiconductor materials. Alternatively or additionally, the substrate **201** includes other elementary semiconductor materials such as germanium, gallium arsenic, or other suitable semiconductor materials. In some embodiments, the substrate **201** may further include other features such as various doped regions, buried layer(s), and/or epitaxy layer(s). Moreover, in some embodiments, the substrate **201** is made of an alloy semiconductor such as silicon germanium, silicon germanium carbide, gallium arsenic phosphide, or gallium indium phosphide. Furthermore, the substrate **201** may be a semiconductor on an insulator such as silicon on insulator (SOI) or silicon on sapphire.

(35) The insulating layer **202**, the ferroelectric layer **230**, and the gate electrode **220** are sequentially stacked on the channel region **210** of the substrate **100**. In some embodiments, the combination of the insulating layer **202** and the ferroelectric layer **230** may be referred to as a gate dielectric layer. The insulating layer **202** may include silicon oxide, silicon nitride, silicon oxynitride, a high-K dielectric material, or the like, or combinations thereof. The high-k dielectric material may have a dielectric constant such as greater than about 4, or greater than about 7 or 10.

(36) In some embodiments, the high-k dielectric material includes ZrO₂, Gd₂O₃, HfO₂, BaTiO₃, Al₂O₃, LaO₂, TiO₂, Ta₂O₅, Y₂O₃, STO, BTO, BaZrO, HfZrO, HfLaO, HfTaO, HfTiO, combinations thereof, or other suitable material. In some embodiments, a native oxide layer (e.g., silicon oxide, not shown) may exist between the insulating layer **202** and the substrate **201**. In some other embodiments, the insulating layer **202** is optionally formed and may be omitted. If the insulating layer **202** is omitted, the ferroelectric memory device **200** may be referred to as a metal-ferroelectric-semiconductor (MFS) ferroelectric memory device.

(37) The gate electrode **220** may include polysilicon and/or metallic materials. In some embodiments, the gate electrode **220** includes a work function metal layer and a metal layer on the work function metal layer. The work function metal layer is configured to tune the work function of the transistor to achieve a desired threshold voltage V_t . The work function metal layer may be an N-type work function metal layer or a P-type work function metal layer. In some embodiments, the P-type work function metal layer includes a metal with a sufficiently large effective work function and may include one or more of the following: TiN, WN, TaN, conductive metal oxide, and/or other suitable material, or combinations thereof. In alternative embodiments, the N-type work function metal layer includes a metal with sufficiently low effective work function and may include one or more of the following: tantalum (Ta), titanium aluminide (TiAl), titanium aluminum nitride (TiAlN), tantalum carbide (TaC), tantalum carbide nitride (TaCN), tantalum silicon nitride (TaSiN), titanium silicon nitride (TiSiN), other suitable metals, suitable conductive metal oxide, or combinations thereof. The metal layer may include copper, aluminum, tungsten, cobalt (Co), or any other suitable metallic material, or the like or combinations thereof.

(38) In the present embodiment, the ferroelectric layer **230** is disposed between channel region **210** (i.e., the first conductive region **110** of FIG. 1) and the gate electrode **220**. The composition and structural features of the ferroelectric layer **230** have been described above in connection with FIG. 1, and are not repeated here.

(39) The S/D regions **205A** and **205B** may be doped regions including p-type dopants, such as boron, BF₂ P, and/or a combination thereof. Alternatively, the S/D regions **205A** and **205B** may be doped regions including n-type dopants, such as phosphorus, arsenic, and/or a combination thereof. In some embodiments, the S/D regions **205A** and **205B** may also include an epitaxial layer (or strained layer).

(40) In some embodiments, the gate stack including the insulating layer **202**, the ferroelectric layer **230**, and the gate electrode **220** are formed by various deposition processes and patterning processes. The S/D regions **205A** and **205B** may be formed by the doping process and/or epitaxial

process.

(41) In some embodiments, an interlayer dielectric layer (ILD) **221** is disposed on the substrate **201** and covering the transistor. An interconnection structure (not shown) may be disposed over the interlayer dielectric layer **221** and electrically coupled to the transistor. In some embodiments, the ILD **221** is formed after the formation of gate stack.

(42) In the ferroelectric memory device **200**, the ferroelectric layer **230** has polarization characteristics in accordance with a voltage applied through the gate electrode **220**, and a conductive region (i.e., the channel region **210**) is formed between the S/D regions **205A** and **205B**. As a result, a current flows between the S/D regions **205A** and **205B**. When the voltage applied through the gate electrode is cut off, the polarization characteristics of the ferroelectric layer **230** is continuously maintained. Therefore, the ferroelectric memory device **200** can function as a nonvolatile memory device.

(43) FIG. 3 is a cross-sectional view illustrating a ferroelectric memory device **300** which is a 1 transistor-1 capacitor (1T1C) type ferroelectric memory device, in accordance with some embodiments. In some embodiments, the ferroelectric memory device **300** in FIG. 3 includes a substrate **301**, a transistor **306** disposed on and/or in the substrate **301** and a ferroelectric tunnel junction **350** electrically connected to the transistor **306**. The ferroelectric tunnel junction **350** includes a ferroelectric layer **330** disposed between a first conductive region **310** and a second conductive region **320**. In some embodiments, the transistor **306** includes a gate dielectric layer **302**, a gate electrode **303**, a channel region **304**, and a source region **305A** and a drain region **305B** in the substrate **301** and on sides of the gate electrode **303**. The ferroelectric tunnel junction **350** is electrically connected to the drain region **305B** of the transistor **306** through a conductive plug/via **309**. In some embodiments, a first ILD **307** is disposed on the substrate **301** and adjacent the gate electrode **303** and the gate dielectric layer **302**. A second ILD **308** is disposed on the first ILD **307** and covering top surface of the gate electrode **303**. The conductive plug **309** penetrates through the ILDs **307** and **308** to electrically connect to the drain region **305B** of the transistor **306**. A dielectric layer **321** may be disposed on the second ILD **308** and laterally adjacent the ferroelectric tunnel junction **350**. In some embodiments, the ferroelectric tunnel junction **350** is embedded in an interconnection structure over the transistor **306**. In some other embodiments, the ferroelectric tunnel junction **350** may be connected to the drain region **305** through more than one conductive plug and/or conductive lines.

(44) FIGS. 2 and 3 illustrate examples of ferroelectric tunnel junction **100** as part of a non-volatile memory device. However, the applications of the ferroelectric tunnel junction **100** described above are merely for illustration, and the disclosure is not limited thereto. For example, in some other embodiments, the ferroelectric tunnel junction **100** may also be applied in other kinds of ferroelectric devices using ferroelectric materials, such as negative capacitance field effect transistors (NCFET).

(45) FIGS. 4-7 illustrate ferroelectric tunnel junctions (FTJ) and the effect of interfacial layers and electrode composition on the performance of the ferroelectric tunnel junctions, in accordance with some embodiments.

(46) FIG. 4 illustrates a ferroelectric tunnel junction and its band diagrams, in accordance with some embodiments. As shown in FIG. 4, a ferroelectric tunnel junction (FTJ) **400** includes a first electrode **410**, a ferroelectric layer **430** disposed over the first electrode **410**, and a second electrode **420** disposed over the ferroelectric layer **430**. The ferroelectric layer includes a ferroelectric material. In some embodiments, the first electrode is also referred to as the bottom electrode (BE), and the second electrode **420** is also referred to as the top electrode (TE). In some embodiments, ferroelectric tunnel junction (FTJ) **400** in FIG. 4 is similar to the ferroelectric tunnel junctions described above in connection with FIGS. 1-3. In some embodiments, the ferroelectric layer **430** includes a ferroelectric material HfZrO. It is understood, however, that other ferroelectric materials can also be used, including those described above in connection with FIGS. 2-3.

(47) In FIG. 4, diagram 450 is an energy band diagram of ferroelectric tunnel junction 400. In FIG. 4, reference numeral 411 indicates the Fermi level of the first electrode 410, reference numeral 421 indicates the Fermi level of the second electrode 420, reference numeral 431 indicates the valence band edge of the ferroelectric layer 430, and reference numeral 432 indicates the conduction band edge of the ferroelectric layer 430. The polarization in the ferroelectric layer 430 produces charge dipoles 434. Further, charges 414 and 424 are induced in the first electrode 410 and the second electrode 420, respectively.

(48) FIG. 5 illustrates another ferroelectric tunnel junction and its band diagrams, in accordance with some embodiments. As shown in FIG. 5, a ferroelectric tunnel junction (FTJ) 500 includes a first electrode 510, a ferroelectric layer 530 disposed over the first electrode 510, and a second electrode 520 disposed over the ferroelectric layer 530. The ferroelectric layer includes a ferroelectric material. Ferroelectric tunnel junction (FTJ) 500 also includes an interfacial layer 540 disposed on the first electrode 510, and the ferroelectric layer 530 is disposed on the interfacial layer 540. In some embodiments, the first electrode is also referred to as the bottom electrode (BE), and the second electrode 520 is also referred to as the top electrode (TE).

(49) In some embodiments, ferroelectric tunnel junction (FTJ) 500 is similar to the ferroelectric tunnel junctions described above in connection with FIGS. 1-4. A difference is that an interfacial layer 540 is present between the first electrode 510 and the ferroelectric layer 530. In some embodiments, the interfacial layer is a native oxide or a chemical oxide that forms during device fabrication in exposure to oxygen and/or moisture, and thermal process conditions such as annealing processes. Further, native oxide can form during ex-situ deposition procedure or due to the presence of interfacial oxide due to oxygen scavenging from electrodes. The electrical characteristic of ferroelectric tunnel junctions is often not well controlled due to the presence of native oxide.

(50) When a non-ferroelectric (or non-polar) layer is present next to the ferroelectric film, FTJ behavior is modified, i.e., tunneling current can be modulated by the polarization of the ferroelectric film. Different location of the non-ferroelectric layer will give different FTJ characteristics. When the non-ferroelectric layer locates in-between the top electrode (where bias is applied) and the ferroelectric layer, negative polarization will give higher tunneling current and positive polarization will give lower tunneling current. The trend is reversed if the non-ferroelectric layer locates in-between the bottom electrode and the ferroelectric layer.

(51) Since native oxides can be difficult to control, the method of forming a ferroelectric tunnel junction sometimes includes an additional deposition step to deposit layer of a few atomic layers to form a controlled interfacial layer of non-ferroelectric material. If the non-ferroelectric layer is too thick, it could depolarize the ferroelectric layer, and the FTJ function cannot be produced. On the other hand, it is difficult to form a very thin interfacial layer in a controlled manner using related processes.

(52) Some embodiments of this disclosure provide a method of forming the interfacial layer in a ferroelectric tunnel junction by controlling the nitrogen percentage in a nitrogen-containing electrode. In this case, an interfacial layer is formed due to oxygen scavenging from electrodes. The benefits provided by this method include the ability to form a controllable interfacial layer between an electrode and the ferroelectric layer and to selectively form the interfacial layer adjacent to one of electrodes to enhance the device performance of the ferroelectric tunnel junction. Further advantages of the method include compatibility with the CMOS process and cost saving by skipping a deposition step to control the interfacial layer with high precision requirements in the related process. The method and device structures are described below with reference to FIGS. 5-8.

(53) FIG. 5 also shows the effect of the presence of the interfacial layer 540 on the band diagram of the ferroelectric tunnel junction 500. In FIG. 5, diagram 550 is an energy band diagram of ferroelectric tunnel junction 500 with a positive bias voltage on the first electrode 510. In FIG. 5, reference numeral 511 indicates the Fermi level of the first electrode 510, reference numeral 521

indicates the Fermi level of the second electrode **520**, reference numeral **531** indicates the valence band edge of the ferroelectric layer **530**, and reference numeral **532** indicates the conduction band edge of the ferroelectric layer **530**. The polarization in the ferroelectric layer **530** produces charge dipoles **534**. Reference numeral **541** indicates the valence band edge of the interfacial layer **540**, and reference numeral **542** indicates the conduction band edge of the interfacial layer **540**. The polarization in the ferroelectric layer **530** induces charge **544** in the interfacial layer **540**. Further, charges **514** and **524** are induced in the first electrode **510** and the second electrode **520**, respectively.

(54) Diagram **550** shows band bending and an energy barrier **551** with a positive bias voltage on the first electrode **510**. In contrast, diagram **560** is an energy band diagram of ferroelectric tunnel junction **500** with a positive bias voltage on the second electrode **520**. The resultant energy barrier is shown as **561**.

(55) FIG. **6** illustrates another ferroelectric tunnel junction and its band diagram, in accordance with some embodiments. As shown in FIG. **6**, a ferroelectric tunnel junction (FTJ) **600** includes a first electrode **610**, a ferroelectric layer **630** disposed over the first electrode **610**, and a second electrode **620** disposed over the ferroelectric layer **630**. The ferroelectric layer includes a ferroelectric material. Ferroelectric tunnel junction (FTJ) **600** also includes an interfacial layer **640** disposed on the first electrode **610**, and the ferroelectric layer **630** is disposed on the interfacial layer **640**. In some embodiments, the first electrode is also referred to as the bottom electrode (BE), and the second electrode **620** is also referred to as the top electrode (TE).

(56) In some embodiments, ferroelectric tunnel junction (FTJ) **600** is similar to the ferroelectric tunnel junctions described above in connection with FIGS. **1-5** and can be formed using similar materials and processes described above. In FIG. **6**, ferroelectric tunnel junction **600** includes a first nitrogen-containing electrode **610**, characterized by a first nitrogen percentage, shown in diagram **651**. Ferroelectric tunnel junction **600** also includes a ferroelectric layer **630** disposed over the first nitrogen-containing electrode **610**, and the ferroelectric layer **630** includes one or more ferroelectric materials. Ferroelectric tunnel junction **600** also has a second nitrogen-containing electrode **620** disposed over the ferroelectric layer **630**. The second nitrogen-containing electrode **620** is characterized by a second nitrogen percentage, shown in diagram **652**.

(57) In some embodiments, the first nitrogen-containing electrode **610** includes titanium nitride (TiN) or tantalum nitride (TaN), and the second nitrogen-containing electrode **620** also includes titanium nitride (TiN) or tantalum nitride (TaN). In some embodiments, the TiN is characterized by a percentage of nitrogen of 20%-65%. In some embodiments, TaN is characterized by a percentage of nitrogen of 40%-60%. The nitrogen-containing electrodes are not limited to TiN or TaN, and can include suitable metal nitrides or other nitrogen-containing conductors, or the combination thereof.

(58) In FIG. **6**, the presence of the interfacial layer **640** affects the band diagram of the ferroelectric tunnel junction **600**, similarly to the description of the band diagrams of ferroelectric tunnel junction (FTJ) **500** in connection to FIG. **5**.

(59) In some embodiments, the interfacial layer **640** is formed during the process of forming the ferroelectric tunnel junction **600**, due to the scavenging of oxygen from the ferroelectric layer **630** by the electrodes. In some embodiments, the first nitrogen percentage is less than the second nitrogen percentage, and the interfacial layer is formed between the first nitrogen-containing electrode **610** and the interfacial layer **640**.

(60) For example, as shown in diagram **650** in FIG. **6**, the first nitrogen-containing electrode **610** is characterized by a graded nitrogen profile **651** that increases as it extends away from an interface **640-1** between the interfacial layer **640** and the first nitrogen-containing electrode **610**, starting with a nitrogen percentage **651-1** at the interface **640-1** between the first interfacial layer and the first nitrogen-containing electrode **610** and reaches a higher nitrogen percentage **651-2** at the other surface of the first nitrogen-containing electrode **610**. As shown in diagram **650**, nitrogen percentage **651-1** is lower than the second nitrogen percentage **652-1** in the second nitrogen-

containing electrode **620**. In this example, nitrogen percentage **651-1** is about 20% and **651-2** is about 70%, and nitrogen percentage **652-1** is about 50%. However, other suitable percentages can also be used.

(61) In FIG. **6**, diagram **660** illustrates the magnitude of tunneling current versus applied voltage, in accordance with some disclosure. The band diagrams for ferroelectric tunnel junction **600** are similar to those for ferroelectric tunnel junction **500** in FIG. **5**, where the ferroelectric layer **540** is located adjacent to the first electrode **510**. In this case, the energy barrier **551** is lower than the energy barrier **561**. As shown in diagram **660**, the ferroelectric tunnel junction **600** shows larger tunneling current **661** when a positive remnant polarization +Pr is stored, and lower tunneling current **662** when a negative remnant polarization -Pr is stored. Thus, FIG. **6** illustrates an example of tuning the device performance of a ferroelectric tunnel junction by adjusting the nitrogen percentage in the nitrogen-containing electrodes.

(62) FIG. **7** illustrates another ferroelectric tunnel junction and its band diagram, in accordance with some embodiments. As shown in FIG. **7**, a ferroelectric tunnel junction (FTJ) **700** includes a first electrode **710**, a ferroelectric layer **730** disposed over the first electrode **710**, and a second electrode **720** disposed over the ferroelectric layer **730**. The ferroelectric layer includes a ferroelectric material. Ferroelectric tunnel junction (FTJ) **700** also includes an interfacial layer **740** disposed on the ferroelectric layer **730**, and the second electrode **720** is disposed on the interfacial layer **740**. As noted above, the first electrode is also referred to as the bottom electrode (BE), and the second electrode **720** is also referred to as the top electrode (TE).

(63) In some embodiments, ferroelectric tunnel junction (FTJ) **700** is similar to the ferroelectric tunnel junctions described above in connection with FIGS. **1-6** and can be formed using similar materials and processes described above. In FIG. **7**, ferroelectric tunnel junction **700** includes a first nitrogen-containing electrode **710**, characterized by a first nitrogen percentage, shown in diagram **751**. Ferroelectric tunnel junction **700** also includes a ferroelectric layer **730** disposed over the first nitrogen-containing electrode **710**, and the ferroelectric layer **730** includes one or more ferroelectric materials. Ferroelectric tunnel junction **700** also has a second nitrogen-containing electrode **720** disposed over the interfacial layer **740**. The second nitrogen-containing electrode **720** is characterized by a second nitrogen percentage, shown in diagram **752**.

(64) In some embodiments, the first nitrogen-containing electrode **710** includes titanium nitride (TiN) or tantalum nitride (TaN), and the second nitrogen-containing electrode **720** also includes titanium nitride (TiN) or tantalum nitride (TaN). In some embodiments, the TiN is characterized by a percentage of nitrogen of 20%-65%. In some embodiments, TaN is characterized by a percentage of nitrogen of 40%-60%. The nitrogen-containing electrodes are not limited to TiN or TaN, and can include suitable metal nitrides or other nitrogen-containing conductors, or the combination thereof.

(65) In FIG. **7**, the presence of the interfacial layer **740** affects the band diagram of the ferroelectric tunnel junction **700**, similarly to the description of the band diagrams of ferroelectric tunnel junction (FTJ) **500** in connection to FIG. **5**, but with the positions of the ferroelectric layer **730** and the interfacial layer **740** interchanged.

(66) In some embodiments, the interfacial layer **740** is formed during the process of forming the ferroelectric tunnel junction **700**, due to the scavenging of oxygen from the ferroelectric layer **730** by the electrodes. In some embodiments, the first nitrogen percentage is higher than the second nitrogen percentage, and the interfacial layer is formed between the ferroelectric layer **730** and the second nitrogen-containing electrode **720**.

(67) For example, as shown in diagram **750** in FIG. **7**, the second nitrogen-containing electrode **720** is characterized by a graded nitrogen profile **752** that increases as it extends away from an interface between the interfacial layer **740** and the second nitrogen-containing electrode **720**, starting with a nitrogen percentage **752-1** at the interface **740-1** between the interfacial layer **740** and the second nitrogen-containing electrode **720** and reaches a higher nitrogen percentage **752-2** at the other surface of the second nitrogen-containing electrode **720**. As shown in diagram **750**, nitrogen

percentage **752-1** is lower than the second nitrogen percentage **751-1** in the first nitrogen-containing electrode **710**.

(68) In FIG. 7, diagram **760** illustrates the magnitude of tunneling current versus applied voltage, in accordance with some disclosures. The band diagrams for ferroelectric tunnel junction **700** is similar to those for ferroelectric tunnel junction **500** in FIG. 5, with the locations of the interfacial layer and the ferroelectric layer interchanged. As shown in diagram **660**, the ferroelectric tunnel junction **700** shows larger tunneling current **762** when negative remnant polarization $-Pr$ is stored, and lower tunneling current **761** when a positive remnant polarization $+Pr$ is stored. Thus, FIG. 7 illustrates another example of tuning the device performance of a ferroelectric tunnel junction by adjusting the nitrogen percentage in the nitrogen-containing electrodes.

(69) FIG. 8 is a flowchart illustrating a method for forming a semiconductor device, in accordance with some embodiments. As shown in FIG. 8, the method **800** includes a process for forming a ferroelectric tunnel junction (FTJ). The process includes: At **810**—forming a first nitrogen-containing electrode on a substrate, the first nitrogen-containing electrode characterized by a first nitrogen percentage; At **820**—optionally, forming a first interfacial layer on the first nitrogen-containing electrode; At **830**—forming a ferroelectric layer over the first nitrogen-containing electrode, the ferroelectric layer comprising a ferroelectric material; At **840**—optionally, forming a second interfacial layer on the ferroelectric layer; and At **850**—forming a second nitrogen-containing electrode over the ferroelectric layer, the second nitrogen-containing electrode characterized by a second nitrogen percentage, wherein the first nitrogen percentage is different from the second nitrogen percentage.

(70) The method **800** outlined above is described in connection to FIGS. 6-7. If, as shown in FIG. 6, the first nitrogen percentage **651** is less than the second nitrogen percentage **652**, the method further includes forming an interfacial layer between the first nitrogen-containing electrode and the ferroelectric layer [**630**]. If, as shown in FIG. 7, the first nitrogen percentage is greater than the second nitrogen percentage [**752**], the method further includes forming an interfacial layer between the ferroelectric layer and the second nitrogen-containing electrode [**720**].

(71) In some embodiments, both the first interfacial layer and the second interfacial layer are formed in the same ferroelectric tunnel junction. As noted above, the interfacial layer can include native oxide and/or chemical oxide, besides the oxides formed from oxygen scavenging by the nitrogen-containing electrodes. In some embodiments, the top electrode and the bottom electrodes are characterized by different oxygen percentages to create an asymmetrical charge screening effect at the two interfaces of the ferroelectric layer with the two nitrogen-containing electrodes. The presence of a non-polar material between the polar material and its neighboring electrode results in incomplete charge screening at the interface. The net charge and, hence, band bending and its resulting tunneling current, will change based on the polarization of the polar material. A thicker interfacial layer gives larger band bending and creates a larger difference between tunneling current at $+Pr$ and $-Pr$ cases. The upper boundary of the interfacial layer thickness is the depolarization field. In some embodiments, the thickness of the interfacial layer is controlled within 1 nm.

(72) In some embodiments, as shown in FIG. 7, when the first nitrogen percentage is greater than the second nitrogen percentage, the first interfacial layer is thinner than the second interfacial layer. In this case, the tunneling current is higher with a negative remnant polarization $-Pr$ than a positive remnant polarization $+Pr$. In other embodiments, as shown in FIG. 6, when the first nitrogen percentage is lower than the second nitrogen percentage, the first interfacial layer is thicker than the second interfacial layer. In this case, the tunneling current is higher with a positive remnant polarization $+Pr$ than a negative remnant polarization $-Pr$. In some embodiments, an electrode with a lower nitrogen percentage will produce a thicker interfacial layer compared with the same electrode with a higher nitrogen percentage.

(73) In some embodiments, the first nitrogen-containing electrode and the second nitrogen-containing electrode each includes titanium nitride (TiN) or tantalum nitride (TaN). In some

embodiments, the TiN is characterized by a percentage of nitrogen of 20%-65%, and the TaN is characterized by a percentage of nitrogen of 40%-60%.

(74) In some embodiments of method **800**, the first nitrogen-containing electrode is characterized by a graded nitrogen profile that increases as it extends away from an interface between the first interfacial layer and the first nitrogen-containing electrode, starting with a nitrogen percentage at the interface between the first interfacial layer and the first nitrogen-containing electrode that is lower than the second nitrogen percentage in the second nitrogen-containing electrode.

(75) In some other embodiments of method **800**, the second nitrogen-containing electrode is characterized by a graded nitrogen profile that increases as it extends away from an interface between the second interfacial layer and the second nitrogen-containing electrode, starting with a nitrogen percentage at the interface between the second interfacial layer and the second nitrogen-containing electrode that is lower than the first nitrogen percentage in the first nitrogen-containing electrode.

(76) In some embodiments, the nitrogen-containing electrode, such as TiN and TaN, is formed by sputtering or physical vapor deposition, using argon (Ar₂) as the carrier stabilization gas and nitrogen (N₂) as the reactive gas. The nitrogen percentage in the electrode can be varied by changing the flow rate of the nitrogen gas.

(77) In some embodiments, the nitrogen percentage profiles can be uniform for both electrodes, with one lower than the other. However, a uniform low nitrogen profile through the thickness of the electrode is undesirable, because it may cause overly strong oxygen scavenging, leading to deterioration or breakdown of the ferroelectric material.

(78) In some embodiments, as shown in FIG. 3, the method also includes forming a transistor having a gate region, drain region, and source region, and forming an interconnect structure to connect the ferroelectric tunnel junction to the drain region of the transistor to form a memory device.

(79) In some embodiments, as shown in FIG. 2, the method also forms a transistor that includes: a source region, a drain region, a channel region disposed between the source region and the drain region, and a gate structure disposed on the channel region. The gate structure includes the ferroelectric tunnel junction described above and a gate metal layer disposed over the ferroelectric tunnel junction.

(80) Additional embodiments provide methods and devices that vary the oxygen content of oxygen-containing electrodes to enhance the device performance of ferroelectric tunnel junctions. These embodiments are described below with reference to FIGS. 9-15

(81) FIG. 9 illustrates another ferroelectric tunnel junction and its band diagram, in accordance with some embodiments. As shown in FIG. 9, a ferroelectric tunnel junction (FTJ) **900** includes a first electrode **910**, a ferroelectric layer **930** disposed over the first electrode **910**, and a second electrode **920** disposed over the ferroelectric layer **930**. The ferroelectric layer includes a ferroelectric material. Ferroelectric tunnel junction (FTJ) **900** also includes an interfacial layer **940** disposed on the first electrode **910**, and the ferroelectric layer **930** is disposed on the interfacial layer **940**. In some embodiments, the first electrode **910** is also referred to as the bottom electrode (BE), and the second electrode **920** is also referred to as the top electrode (TE).

(82) In some embodiments, ferroelectric tunnel junction (FTJ) **900** is similar to the ferroelectric tunnel junctions **400-800** described above in connection with FIGS. 4-8. One difference is that in ferroelectric tunnel junction (FTJ) **900**, the electrodes are oxygen-containing electrodes. One example of the oxygen-containing electrodes is the transparent conductive oxides, also referred to as TCO, for example, ZnO, In₂O₃, and SnO₂, AZO, TCO, and ITO, etc. These transparent conductive oxides have work functions from 3.0 eV to 5.4 eV, and their work functions can be varied by varying the oxygen content in the fabrication process.

(83) In FIG. 9, diagram **950** is an energy band diagram of ferroelectric tunnel junction **900**. In FIG. 9, reference numeral **911** indicates the Fermi level of the first electrode **910**, reference numeral **921**

indicates the Fermi level of the second electrode **920**, reference numeral **931** indicates the valence band edge of the ferroelectric layer **930**, and reference numeral **932** indicates the conduction band edge of the ferroelectric layer **930**. Reference numeral **941** indicates the valence band edge of the interfacial layer **940**, and reference numeral **942** indicates the conduction band edge of the interfacial layer **940**.

(84) Note that diagram **950** shows band bending and an energy barrier **961** determined by the work functions of the first electrode **910** and the second electrode **920**. FIG. **9** also shows that if the work functions can be reduced, e.g., as indicated by Fermi levels **912** and **922**, then the energy barrier **961** will become smaller, leading to higher tunneling current.

(85) In some embodiments, the work function of the TCO electrode is adjusted by the oxygen content to enhance the tunneling current of the ferroelectric tunnel junction. FIG. **10** is a table illustrating possible oxygen profiles in a ferroelectric tunnel junction, depending on the location of the interfacial layer, in accordance with some embodiments. In FIG. **10**, the interfacial layer is labeled “Non-polar layer.” Four examples are shown. Example 1: The interfacial layer is located adjacent to the top electrode, the oxygen content of the top electrode is 10%~15% and the oxygen content of the bottom electrode is 25%~30%. This example is illustrated in FIG. **11**. Example 2: The interfacial layer is located adjacent to the top electrode, the oxygen content of the top electrode is 25%~30% and the oxygen content of the bottom electrode is 10%~15%. This example is illustrated in FIG. **12**. Example 3: The interfacial layer is located adjacent to the bottom electrode, the oxygen content of the top electrode is 10%~15% and the oxygen content of the bottom electrode is 25%~30%. This example is illustrated in FIG. **13**. Example 4: The interfacial layer is located adjacent to the bottom electrode, the oxygen content of the top electrode is 25%~30% and the oxygen content of the bottom electrode is 10%~15%. This example is illustrated in FIG. **14**.

(86) FIG. **11** illustrates another ferroelectric tunnel junction and its band diagram, in accordance with some embodiments. As shown in FIG. **11**, a ferroelectric tunnel junction (FTJ) **1100** includes a first electrode **1110**, a ferroelectric layer **1130** disposed over the first electrode **1110**, and a second electrode **1120** disposed over the ferroelectric layer **1130**. The ferroelectric layer includes a ferroelectric material. Ferroelectric tunnel junction (FTJ) **1100** also includes an interfacial layer **1140** disposed on the ferroelectric layer **1130**, and the second electrode **1120** is disposed on the interfacial layer **1140**. In some embodiments, the first electrode **1110** is also referred to as the bottom electrode (BE), and the second electrode **1120** is also referred to as the top electrode (TE).

(87) In the example of FIG. **11**, the first electrode **1110** is an oxygen-containing electrode characterized by a first oxygen percentage, and the second electrode **1120** is an oxygen-containing electrode characterized by a second oxygen percentage. In this example, the oxygen-containing electrode is TCO. However, it is understood that the disclosure applies to other oxygen-containing electrodes as well.

(88) Diagrams **1150A** and **1150B** illustrate the band diagrams for ferroelectric tunnel junction **1100** with a negative remnant polarization $-Pr$ and a positive remnant polarization $+Pr$, respectively, when both electrodes are characterized by the same oxygen percentage, for example, 25%~30%. In diagram **1150A**, with the ferroelectric tunnel junction **1100** with a negative remnant polarization $-Pr$, the label **1111** indicates the Fermi level of the first electrode **1110**, the label **1121** indicates the Fermi level of the second electrode **1120**, the label **1131** indicates the band diagram of the ferroelectric layer **1130**, and the label **1141** indicates the band diagram of the interfacial layer **1140**. In diagram **1150A**, label **1151** indicates the work function of the second electrode **1120** with respect to the interfacial layer **1140**. Similarly, diagram **1150B** illustrates the band diagram for ferroelectric tunnel junction **1100** with a positive remnant polarization $+Pr$, and the label **1152** indicates the work function of the second electrode **1120** with respect to the interfacial layer **1140**.

(89) Next, the effect of different oxygen percentages in the electrodes are considered. The first electrode **1110** is characterized by a first oxygen percentage, for example, 25%~30%, as illustrated in diagram **1161**. The second electrode **1120** is characterized by a second oxygen percentage, for

example, 10%~15%, as illustrated in diagram **1162**. Diagrams **1170A** and **1170B** illustrate the band diagram for ferroelectric tunnel junction **1100** with a negative remnant polarization $-Pr$ and a positive remnant polarization $+Pr$, respectively. In diagram **1150A**, with the ferroelectric tunnel junction **1100** with a negative remnant polarization $-Pr$, the label **1111** indicates the Fermi level of the first electrode **1110**, the label **1121** indicates the Fermi level of the second electrode **1120**, the label **1131** indicates the band diagram of the ferroelectric layer **1130**, and the label **1141** indicates the band diagram of the interfacial layer **1140**. In diagram **1170A**, label **1171** indicates the work function of the second electrode **1120** with respect to the interfacial layer **1140**. Similarly, diagram **1170B** illustrates the band diagram for ferroelectric tunnel junction **1100** with a positive remnant polarization $+Pr$, and the label **1172** indicates the work function of the second electrode **1120** with respect to the interfacial layer **1140**.

(90) It can be seen that, when the interfacial layer **1140** is disposed adjacent to the second electrode **1120**, and the second electrode **1120** has a lower oxygen percentage than the first electrode **1110**, the work function of the second electrode **1120** is lowered. For example, the work function **1171** is lower than work function **1151**, and work function **1172** is lower than work function **1152**. As a result, the tunneling current level is enhanced for positive tunneling current under both negative remnant polarization $-Pr$ and positive remnant polarization $+Pr$.

(91) FIG. **12** illustrates another ferroelectric tunnel junction and its band diagram, in accordance with some embodiments. As shown in FIG. **12**, a ferroelectric tunnel junction (FTJ) **1200** includes a first electrode **1210**, a ferroelectric layer **1230** disposed over the first electrode **1210**, and a second electrode **1220** disposed over the ferroelectric layer **1230**. The ferroelectric layer includes a ferroelectric material. Ferroelectric tunnel junction (FTJ) **1200** also includes an interfacial layer **1240** disposed on the first electrode **1210**, and the ferroelectric layer **1230** is disposed on the interfacial layer **1240**. In some embodiments, the first electrode **1210** is also referred to as the bottom electrode (BE), and the second electrode **1220** is also referred to as the top electrode (TE).

(92) In the example of FIG. **12**, the first electrode **1210** is an oxygen-containing electrode characterized by a first oxygen percentage, and the second electrode **1220** is an oxygen-containing electrode characterized by a second oxygen percentage. In this example, the oxygen-containing electrode is TCO, although the disclosure applies to other oxygen-containing electrodes as well.

(93) Diagrams **1250A** and **1250B** illustrate the band diagrams for ferroelectric tunnel junction **1200** with a negative remnant polarization $-Pr$ and a positive remnant polarization $+Pr$, respectively, when both electrodes are characterized by the same oxygen percentage, for example, 25%~30%. In diagram **1250A**, with the ferroelectric tunnel junction **1200** with a negative remnant polarization $-Pr$, the label **1211** indicates the Fermi level of the first electrode **1210**, the label **1221** indicates the Fermi level of the second electrode **1220**, **1231** indicates the band diagram of the ferroelectric layer **1230**, and **1241** indicates the band diagram of the interfacial layer **1240**. In diagram **1250A**, label **1251** indicates the work function of the first electrode **1210** with respect to the interfacial layer **1240**. Similarly, diagram **1250B** illustrates the band diagram for ferroelectric tunnel junction **1200** with a positive remnant polarization $+Pr$, and the label **1252** indicates the work function of the first electrode **1210** with respect to the interfacial layer **1240**.

(94) Next, the effect of different oxygen percentages in the electrodes is considered. The first electrode **1210** is characterized by a first oxygen percentage, for example, 10%~15%, as illustrated in diagram **1261**. The second electrode **1220** is characterized by a second oxygen percentage, for example, 25%~30%, as illustrated in diagram **1262**. Diagrams **1270A** and **1270B** illustrate the band diagram for ferroelectric tunnel junction **1200** with a negative remnant polarization $-Pr$ and a positive remnant polarization $+Pr$, respectively. In diagram **1270A**, with the ferroelectric tunnel junction **1200** with a negative remnant polarization $-Pr$, the label **1211** indicates the Fermi level of the first electrode **1210**, the label **1221** indicates the Fermi level of the second electrode **1220**, the label **1231** indicates the band diagram of the ferroelectric layer **1230**, and the label **1241** indicates the band diagram of the interfacial layer **1240**. In diagram **1270A**, label **1271** indicates the work

function of the first electrode **1210** with respect to the interfacial layer **1240**. Similarly, diagram **1270B** illustrates the band diagram for ferroelectric tunnel junction **1200** with a positive remnant polarization +Pr, and the label **1272** indicates the work function of the first electrode **1210** with respect to the interfacial layer **1240**.

(95) It can be seen that, when the interfacial layer **1240** is disposed adjacent to the first electrode **1210**, and the second electrode **1220** has a higher oxygen percentage than the first electrode **1210**, the work function of the first electrode **1210** is lowered. For example, the work function **1271** is lower than work function **1251**, and the work function **1272** is lower than work function **1252**. As a result, the tunneling current level is enhanced for negative tunneling current under both negative remnant polarization -Pr and positive remnant polarization +Pr.

(96) FIG. **13** illustrates another ferroelectric tunnel junction and its band diagram, in accordance with some embodiments. As shown in FIG. **13**, a ferroelectric tunnel junction (FTJ) **1300** includes a first electrode **1310**, a ferroelectric layer **1330** disposed over the first electrode **1310**, and a second electrode **1320** disposed over the ferroelectric layer **1330**. The ferroelectric layer includes a ferroelectric material. Ferroelectric tunnel junction (FTJ) **1300** also includes an interfacial layer **1340** disposed on the first electrode **1310**, and the ferroelectric layer **1330** is disposed on the interfacial layer **1340**. In some embodiments, the first electrode **1310** is also referred to as the bottom electrode (BE), and the second electrode **1320** is also referred to as the top electrode (TE).

(97) In the example of FIG. **13**, the first electrode **1310** is an oxygen-containing electrode characterized by a first oxygen percentage, and the second electrode **1320** is an oxygen-containing electrode characterized by a second oxygen percentage. In this example, the oxygen-containing electrode is TCO, although the disclosure applies to other oxygen-containing electrodes as well.

(98) Diagrams **1350A** and **1350B** illustrate the band diagrams for ferroelectric tunnel junction **1300** with a negative remnant polarization -Pr and a positive remnant polarization +Pr, respectively, when both electrodes are characterized by the same oxygen percentage, for example, 25%~30%. In diagram **1350A**, with the ferroelectric tunnel junction **1300** with a negative remnant polarization -Pr, the label **1311** indicates the Fermi level of the first electrode **1310**, the label **1321** indicates the Fermi level of the second electrode **1320**, the label **1331** indicates the band diagram of the ferroelectric layer **1330**, and the label **1341** indicates the band diagram of the interfacial layer **1340**. In diagram **1350A**, label **1351** indicates the work function of the second electrode **1320** with respect to the ferroelectric layer **1330**. Similarly, diagram **1350B** illustrates the band diagram for ferroelectric tunnel junction **1300** with a positive remnant polarization +Pr, and the label **1352** indicates the work function of the second electrode **1320** with respect to the ferroelectric layer **1330**.

(99) Next, the effect of different oxygen percentages in the electrodes is considered. The first electrode **1310** is characterized by a first oxygen percentage, for example, 25%~30%, as illustrated in diagram **1361**. The second electrode **1320** is characterized by a second oxygen percentage, for example, 10%~15%, as illustrated in diagram **1362**. Diagrams **1370A** and **1370B** illustrate the band diagram for ferroelectric tunnel junction **1300** with a negative remnant polarization -Pr and a positive remnant polarization +Pr, respectively. In diagram **1350A**, with the ferroelectric tunnel junction **1300** with a negative remnant polarization -Pr, the label **1311** indicates the Fermi level of the first electrode **1310**, the label **1321** indicates the Fermi level of the second electrode **1320**, the label **1331** indicates the band diagram of the ferroelectric layer **1330**, and the label **1341** indicates the band diagram of the interfacial layer **1340**. In diagram **1370A**, label **1371** indicates the work function of the second electrode **1320** with respect to the ferroelectric layer **1330**. Similarly, diagram **1170B** illustrates the band diagram for ferroelectric tunnel junction **1300** with a positive remnant polarization +Pr, and the label **1372** indicates the work function of the second electrode **1320** with respect to the ferroelectric layer **1330**.

(100) It can be seen that, when the interfacial layer **1340** is disposed adjacent to the first electrode **1310**, and the second electrode **1320** has a lower oxygen percentage than the first electrode **1310**,

the work function of the second electrode **1320** is lowered. For example, the work function **1371** is lower than work function **1351**, and the work function **1372** is lower than work function **1352**. As a result, the tunneling current level is enhanced for positive tunneling current under both negative remnant polarization $-Pr$ and positive remnant polarization $+Pr$.

(101) FIG. **14** illustrates another ferroelectric tunnel junction and its band diagrams, in accordance with some embodiments. As shown in FIG. **14**, a ferroelectric tunnel junction (FTJ) **1400** includes a first electrode **1410**, a ferroelectric layer **1430** disposed over the first electrode **1410**, and a second electrode **1420** disposed over the ferroelectric layer **1430**. The ferroelectric layer includes a ferroelectric material. Ferroelectric tunnel junction (FTJ) **1400** also includes an interfacial layer **1440** disposed on the first electrode **1410**, and the ferroelectric layer **1430** is disposed on the interfacial layer **1440**. In some embodiments, the first electrode **1410** is also referred to as the bottom electrode (BE), and the second electrode **1420** is also referred to as the top electrode (TE).

(102) In the example of FIG. **14**, the first electrode **1410** is an oxygen-containing electrode characterized by a first oxygen percentage, and the second electrode **1420** is an oxygen-containing electrode characterized by a second oxygen percentage. In this example, the oxygen-containing electrode is TCO, although the disclosure applies to other oxygen-containing electrodes as well.

(103) Diagrams **1450A** and **1450B** illustrate the band diagrams for ferroelectric tunnel junction **1400** with a negative remnant polarization $-Pr$ and a positive remnant polarization $+Pr$, respectively, when both electrodes are characterized by the same oxygen percentage, for example, 25%~30%. In diagram **1450A**, with the ferroelectric tunnel junction **1400** with a negative remnant polarization $-Pr$, the label **1411** indicates the Fermi level of the first electrode **1410**, the label **1421** indicates the Fermi level of the second electrode **1420**, the label **1431** indicates the band diagram of the ferroelectric layer **1430**, and the label **1441** indicates the band diagram of the interfacial layer **1440**. In diagram **1450A**, label **1451** indicates the work function of the first electrode **1410** with respect to the interfacial layer **1440**. Similarly, diagram **1450B** illustrates the band diagram for ferroelectric tunnel junction **1400** with a positive remnant polarization $+Pr$, and the label **1452** indicates the work function of the first electrode **1410** with respect to the interfacial layer **1440**.

(104) Next, the effect of different oxygen percentages in the electrodes are considered. The first electrode **1410** is characterized by a first oxygen percentage, for example, 25%~30%, as illustrated in diagram **1461**. The second electrode **1420** is characterized by a second oxygen percentage, for example, 10%~15%, as illustrated in diagram **1462**. Diagrams **1470A** and **1470B** illustrate the band diagram for ferroelectric tunnel junction **1400** with a negative remnant polarization $-Pr$ and a positive remnant polarization $+Pr$, respectively. In diagram **1470A**, with the ferroelectric tunnel junction **1400** with a negative remnant polarization $-Pr$, the label **1411** indicates the Fermi level of the first electrode **1410**, the label **1421** indicates the Fermi level of the second electrode **1420**, **1431** indicates the band diagram of the ferroelectric layer **1430**, and **1441** indicates the band diagram of the interfacial layer **1440**. In diagram **1470A**, label **1471** indicates the work function of the first electrode **1410** with respect to the interfacial layer **1440**. Similarly, diagram **1470B** illustrates the band diagram for ferroelectric tunnel junction **1400** with a positive remnant polarization $+Pr$, and the label **1472** indicates the work function of the first electrode **1410** with respect to the interfacial layer **1440**.

(105) It can be seen that, when the interfacial layer **1440** is disposed adjacent to the first electrode **1410**, and the second electrode **1420** has a lower oxygen percentage than the first electrode **1410**, the work function of the first electrode **1410** is lowered. For example, the work function **1471** is lower than work function **1451**, and the work function **1472** is lower than work function **1452**. As a result, the tunneling current level is enhanced for negative tunneling current under both negative remnant polarization $-Pr$ and positive remnant polarization $+Pr$.

(106) FIG. **15** is a flowchart illustrating a method for forming a semiconductor device, in accordance with some embodiments. As shown in FIG. **15**, the method **1500** includes a process for forming a ferroelectric tunnel junction (FTJ). The process includes: At **1510**—forming a first

oxygen-containing electrode, characterized by a first oxygen percentage; At **1520**—optionally, forming a first interfacial layer on the first oxygen-containing electrode; At **1530**—forming a ferroelectric layer disposed over the first oxygen-containing electrode, the ferroelectric layer comprising a ferroelectric material; At **1540**—optionally, forming a second interfacial layer on the ferroelectric layer; and At **1550**—forming a second oxygen-containing electrode disposed over the ferroelectric layer, and characterized by a second oxygen percentage; wherein the first oxygen percentage is different from the second oxygen percentage.

(107) In some embodiments, the method includes forming a FTJ with enhanced positive tunneling current under $-Pr$ and $+Pr$ by forming a first interfacial layer between the first oxygen-containing electrode and the ferroelectric layer and a second interfacial layer between the ferroelectric layer and the second oxygen-containing electrode. In some embodiments the first oxygen percentage is 25%-30%, and the second oxygen percentage is 10%-15%.

(108) In some embodiments, the method includes forming a FTJ with enhanced negative tunneling current under $-Pr$ and $+Pr$ by forming a first interfacial layer between the first oxygen-containing electrode and the ferroelectric layer and a second interfacial layer between the ferroelectric layer and the second oxygen-containing electrode. In some embodiments, the first oxygen percentage is 10%-15% and the second oxygen percentage is 25%-30%.

(109) Embodiments of the disclosure provide a semiconductor device including a ferroelectric tunnel junction (FTJ), wherein the ferroelectric tunnel junction includes a first electrode, a ferroelectric layer disposed over the first electrode, and a second electrode disposed over the ferroelectric layer. The first electrode contains nitrogen or oxygen and is characterized by a first percentage of nitrogen or oxygen. The second electrode contains nitrogen or oxygen and is characterized by a second percentage of nitrogen or oxygen. The first percentage is different from the second percentage. The percentage of nitrogen or oxygen is adjusted to enhance the device performance of the ferroelectric tunnel junction. In some embodiments, varying the percentage of nitrogen in nitrogen-containing electrodes can lead to different thickness of interfacial layers adjacent to each electrode, resulting in band bending that enhances tunneling current. In some other embodiments, varying the percentage of oxygen in oxygen-containing electrodes can lead to different work functions of each electrode, resulting in enhancement of tunneling current.

(110) In some embodiments, a semiconductor device includes a ferroelectric tunnel junction (FTJ), which includes a first nitrogen-containing electrode, characterized by a first nitrogen percentage; a ferroelectric layer disposed over the first nitrogen-containing electrode, and a second nitrogen-containing electrode disposed over the ferroelectric layer, the second nitrogen-containing electrode characterized by a second nitrogen percentage. The first nitrogen percentage is different from the second nitrogen percentage.

(111) In some embodiments, a method for forming a semiconductor device includes forming a ferroelectric tunnel junction (FTJ). The process of forming the ferroelectric tunnel junction includes forming a first nitrogen-containing electrode on a substrate, the first nitrogen-containing electrode characterized by a first nitrogen percentage, and optionally, forming a first interfacial layer on the first nitrogen-containing electrode. The method also includes forming a ferroelectric layer over the first nitrogen-containing electrode, and optionally, forming a second interfacial layer on the ferroelectric layer. The method further includes forming a second nitrogen-containing electrode over the ferroelectric layer, the second nitrogen-containing electrode characterized by a second nitrogen percentage. The first nitrogen percentage is different from the second nitrogen percentage. If the first nitrogen percentage is less than the second nitrogen percentage, the method further includes forming the first interfacial layer between the first nitrogen-containing electrode and the ferroelectric layer. If the first nitrogen percentage is greater than the second nitrogen percentage, the method further includes forming the second interfacial layer between the ferroelectric layer and the second nitrogen-containing electrode.

(112) In some embodiments, a method for forming a semiconductor device includes forming a

ferroelectric tunnel junction (FTJ). The process of forming the ferroelectric tunnel junction includes forming a first nitrogen-containing electrode on a substrate, the first nitrogen-containing electrode characterized by a first nitrogen percentage. The method also includes forming a ferroelectric layer over the first nitrogen-containing electrode. The method further includes forming a second nitrogen-containing electrode over the ferroelectric layer, the second nitrogen-containing electrode characterized by a second nitrogen percentage. The first nitrogen percentage is different from the second nitrogen percentage. When the first nitrogen percentage is less than the second nitrogen percentage, the method further includes forming a first interfacial layer between the first nitrogen-containing electrode and the ferroelectric layer. When the first nitrogen percentage is greater than the second nitrogen percentage, the method further includes forming a second interfacial layer between the ferroelectric layer and the second nitrogen-containing electrode.

(113) In some embodiments, a semiconductor device includes a ferroelectric tunnel junction (FTJ), which includes a first oxygen-containing electrode characterized by a first oxygen percentage, a ferroelectric layer disposed over the first oxygen-containing electrode, and a second oxygen-containing electrode disposed over the ferroelectric layer, and characterized by a second oxygen percentage, which is different from the first oxygen percentage.

(114) In some embodiments, a method for forming a semiconductor device includes a process for forming a ferroelectric tunnel junction (FTJ). The process includes forming a first oxygen-containing electrode, characterized by a first oxygen percentage, and optionally, forming a first interfacial layer on the first oxygen-containing electrode. The process also includes forming a ferroelectric layer disposed over the first oxygen-containing electrode, and optionally, forming a second interfacial layer on the ferroelectric layer. The process further includes forming a second oxygen-containing electrode disposed over the ferroelectric layer, and characterized by a second oxygen percentage. The first oxygen percentage is different from the second oxygen percentage.

(115) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device, comprising: a ferroelectric tunnel junction (FTJ), wherein the ferroelectric tunnel junction comprises: a first nitrogen-containing electrode, characterized by a first nitrogen percentage; a ferroelectric layer disposed over the first nitrogen-containing electrode, the ferroelectric layer comprising a ferroelectric material; and a second nitrogen-containing electrode disposed over the ferroelectric layer, the second nitrogen-containing electrode characterized by a second nitrogen percentage; and wherein the first nitrogen percentage is different from the second nitrogen percentage.
2. The semiconductor device of claim 1, wherein the first nitrogen-containing electrode comprises titanium nitride (TiN) or tantalum nitride (TaN), and wherein the second nitrogen-containing electrode comprises titanium nitride (TiN) or tantalum nitride (TaN).
3. The semiconductor device of claim 2, wherein the titanium nitride is characterized by a percentage of nitrogen of 20%-65%.
4. The semiconductor device of claim 2, wherein the tantalum nitride is characterized by a percentage of nitrogen of 40%-60%.
5. The semiconductor device of claim 1, wherein the first nitrogen percentage is less than the

second nitrogen percentage, and the ferroelectric tunnel junction further comprises an interfacial layer between the first nitrogen-containing electrode and the ferroelectric layer.

6. The semiconductor device of claim 5, wherein the first nitrogen-containing electrode is characterized by a graded nitrogen profile that increases as it extends away from an interface between the first interfacial layer and the first nitrogen-containing electrode, starting with a nitrogen percentage at the interface between the first interfacial layer and the first nitrogen-containing electrode that is lower than the second nitrogen percentage in the second nitrogen-containing electrode.

7. The semiconductor device of claim 5, wherein the first nitrogen percentage is greater than the second nitrogen percentage, and the ferroelectric tunnel junction further comprises an interfacial layer between the ferroelectric layer and the first nitrogen-containing electrode.

8. The semiconductor device of claim 7, wherein the second nitrogen-containing electrode is characterized by a graded nitrogen profile that increases as it extends away from an interface between the second interfacial layer and the second nitrogen-containing electrode, starting with a nitrogen percentage at the interface between the second interfacial layer and the second nitrogen-containing electrode that is lower than the first nitrogen percentage in the first nitrogen-containing electrode.

9. The semiconductor device of claim 1, further comprising a transistor having a gate region, drain region, and source region; wherein the ferroelectric tunnel junction is connected to the drain region of the transistor to form a memory device.

10. The semiconductor device of claim 1, further comprising a transistor that includes: a source region; a drain region; a channel region disposed between the source region and the drain region; and a gate structure disposed on the channel region, wherein the gate structure includes: the ferroelectric tunnel junction; and a gate metal layer disposed over the ferroelectric tunnel junction.

11. A semiconductor device, comprising: a ferroelectric tunnel junction (FTJ), wherein the ferroelectric tunnel junction comprises: a first oxygen-containing electrode, characterized by a first oxygen percentage; a ferroelectric layer disposed over the first oxygen-containing electrode, the ferroelectric layer comprising a ferroelectric material; and a second oxygen-containing electrode disposed over the ferroelectric layer, and characterized by a second oxygen percentage; wherein the first oxygen percentage is different from the second oxygen percentage.

12. The semiconductor device of claim 11, further comprising a first interfacial layer between the first oxygen-containing electrode and the ferroelectric layer, and wherein first oxygen percentage is between 25%-30%, and the second oxygen percentage is between 10%-15%.

13. The semiconductor device of claim 11, further comprising a first interfacial layer between the first oxygen-containing electrode and the ferroelectric layer, and wherein first oxygen percentage is between 10%-15% and the second oxygen percentage is between 25%-30%.

14. The semiconductor device of claim 11, further comprising a second interfacial layer between the ferroelectric layer and the second oxygen-containing electrode, and wherein first oxygen percentage is 25%-30%, and the second oxygen percentage is 10%-15%.

15. The semiconductor device of claim 11, further comprising a second interfacial layer between the ferroelectric layer and the second oxygen-containing electrode, and wherein first oxygen percentage is 10%-15% and the second oxygen percentage is 25%-30%.

16. A semiconductor device, comprising: a substrate; a ferroelectric tunnel junction (FTJ) disposed on the substrate, wherein the ferroelectric tunnel junction comprises: a first nitrogen-containing electrode characterized by a first nitrogen percentage; a ferroelectric layer disposed over the first nitrogen-containing electrode, the ferroelectric layer comprising a ferroelectric material; and a second nitrogen-containing electrode disposed over the ferroelectric layer, wherein the second nitrogen-containing electrode is characterized by a second nitrogen percentage different from the first nitrogen percentage.

17. The semiconductor device of claim 16, wherein the first nitrogen-containing electrode

comprises titanium nitride (TiN) or tantalum nitride (TaN), and wherein the second nitrogen-containing electrode comprises titanium nitride (TiN) or tantalum nitride (TaN).

18. The semiconductor device of claim 17, wherein the titanium nitride is characterized by a percentage of nitrogen of 20%-65%.

19. The semiconductor device of claim 17, wherein the tantalum nitride is characterized by a percentage of nitrogen of 40%-60%.

20. The semiconductor device of claim 16, wherein the first nitrogen percentage is less than the second nitrogen percentage, and the ferroelectric tunnel junction further comprises an interfacial layer between the first nitrogen-containing electrode and the ferroelectric layer.
